

Computer Hardware: Homework Week 2:

Flip-Flop Counters

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Exercise 2: 4-Bit Asynchronous Up-Counter

This exercise involves the design of a 4-bit asynchronous (ripple) counter using D Flip-Flops.

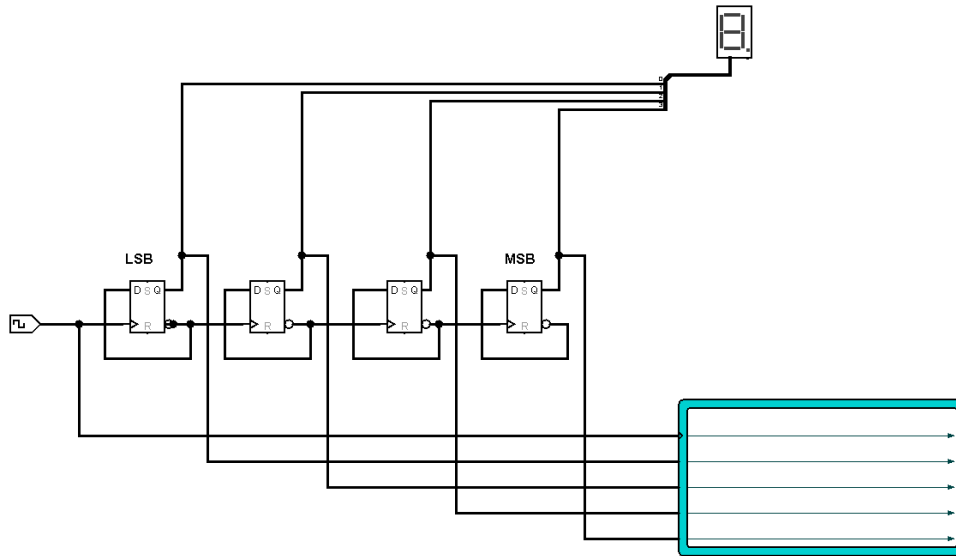


Figure 1: 4-Bit Asynchronous Counter using D Flip-Flops

2.1 Design Logic

To create a toggle effect using a D Flip-Flop, the inverted output $\overline{Q}$ is fed back to the D input. This ensures that on every clock edge, the state flips ($0 \rightarrow 1$ or $1 \rightarrow 0$).

- **Clocking:** The clock signal is only provided to the Least Significant Bit (LSB).
- **Ripple Effect:** Each subsequent flip-flop is clocked by the $\overline{Q}$ output of the preceding stage.

2.2 Output Analysis

The circuit counts from 0000_2 (0) to 1111_2 (15). The output is displayed on a 7-segment hex display, which cycles through 0 – 9 and $A - F$ as the binary value increments.

Exercise 3: Modulo-10 Asynchronous Counter (BCD)

This circuit implements a Modulo-10 counter (decade counter) using JK Flip-Flops, which resets after reaching the count of 9 (1001_2).

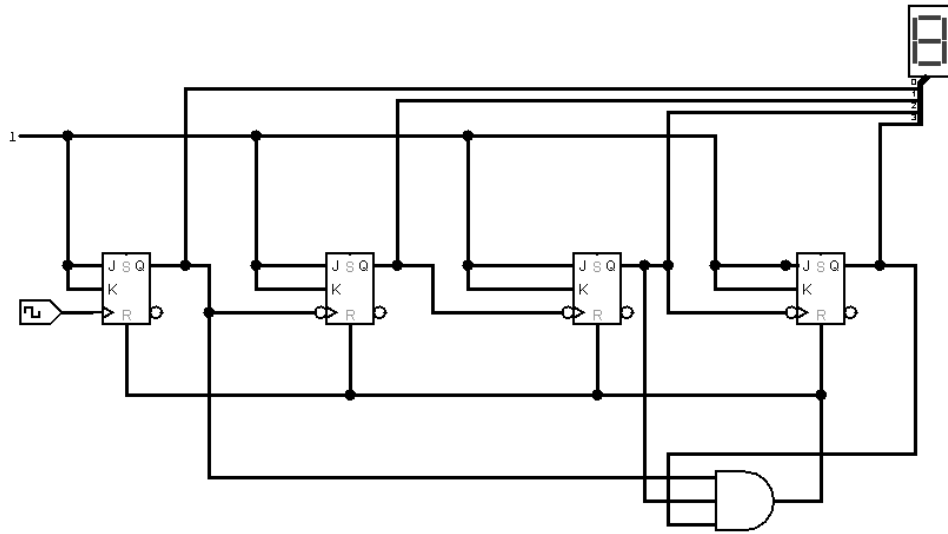


Figure 2: Modulo-10 Counter with Reset Logic

3.1 Reset Mechanism

To truncate the count at 10, an AND gate is used to detect the state 1010_2 .

$$\text{Reset} = Q_3 \cdot Q_1$$

When the counter reaches 10 ($Q_3 = 1, Q_1 = 1$), the AND gate sends a HIGH signal to the asynchronous Clear (CLR) pins of all flip-flops, forcing the count back to 0 immediately.

Exercise 4: 3-Bit Synchronous Counter

This exercise demonstrates a synchronous counter where all flip-flops share a common clock signal to eliminate ripple delays.

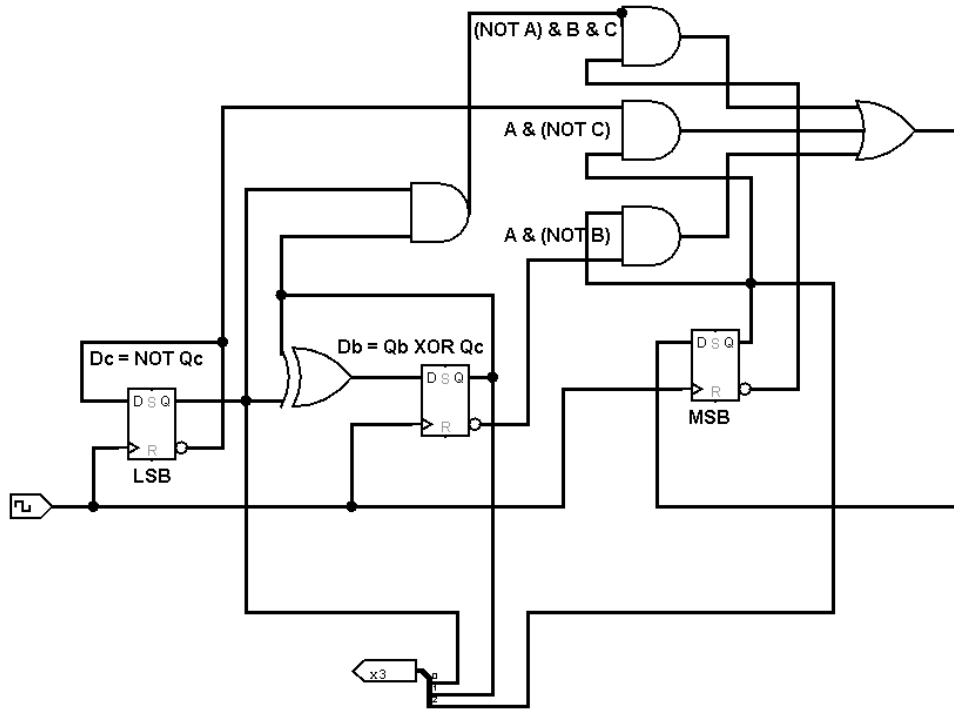


Figure 3: Synchronous Counter Implementation for State Sequence

4.1 Excitation Equations

Based on the logic gates connected to the D inputs in the circuit diagram, the following equations were derived:

- $D_C = \overline{Q_C}$
- $D_B = Q_B \oplus Q_C$
- $D_A = (\overline{Q_A} \cdot Q_B \cdot Q_C) + (Q_A \cdot \overline{Q_C}) + (Q_A \cdot \overline{Q_B})$

4.2 State Transition Table

Applying the present state values ($Q_A Q_B Q_C$) to the excitation equations above yields the next state ($Q_A^+ Q_B^+ Q_C^+$).

Present State			Next State (D Inputs)		
Q_A	Q_B	Q_C	D_A	D_B	D_C
0	0	0	0	0	1
0	0	1	0	1	0
0	1	0	0	1	1
0	1	1	1	0	0
1	0	0	1	0	1
1	0	1	1	1	0
1	1	0	1	1	1
1	1	1	0	0	0

4.3 Karnaugh Maps

The following K-maps are derived from the state transition table to verify the excitation equations for each D flip-flop.

K-map for D_A

Q_A	$Q_B Q_C$			
	00	01	11	10
0	0	0	1	0
1	1	1	0	1

K-map for D_C

Q_A	$Q_B Q_C$			
	00	01	11	10
0	1	0	0	1
1	1	0	0	1

K-map for D_B

Q_A	$Q_B Q_C$			
	00	01	11	10
0	0	1	0	1
1	0	1	0	1

4.4 Verification of Equations

From the K-maps above, we can simplify or verify the logic:

- D_C : The values are 1 when $Q_C = 0$. Thus, $D_C = \overline{Q_C}$.
- D_B : The values follow the XOR pattern for Q_B and Q_C . Thus, $D_B = Q_B \oplus Q_C$.
- D_A : The map confirms the SOP expression: $D_A = (\overline{Q_A}Q_BQ_C) + (Q_A\overline{Q_C}) + (Q_A\overline{Q_B})$.

Exercise 5: Synchronous Counter Using JK Flip-Flops with Direction Control

This exercise implements a 3-bit synchronous counter using JK Flip-Flops. The counter includes a control input C that determines the counting direction:

- $C = 1$: Clockwise (up-counting)
- $C = 0$: Counter-clockwise (down-counting)

All flip-flops are driven by a common clock signal, ensuring synchronous operation.

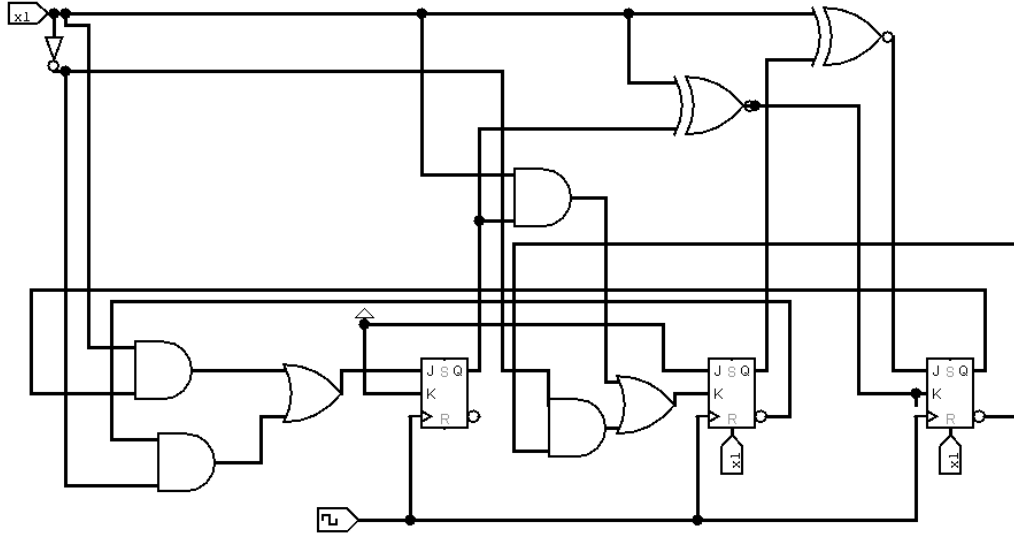


Figure 4: 3-Bit Synchronous Up/Down Counter Using JK Flip-Flops

5.1 Design Principle

A JK Flip-Flop toggles its output when both inputs J and K are set to logic 1. Therefore, the counter is designed by controlling the toggle conditions for each flip-flop stage based on the present state and the control input C .

5.2 Excitation Equations

Let Q_2 , Q_1 , and Q_0 represent the most significant bit (MSB) to the least significant bit (LSB), respectively.

- **LSB (Q_0):** Toggles on every clock cycle

$$J_0 = K_0 = 1$$

- **Middle Bit (Q_1):** Toggles when $Q_0 = 1$ for up-counting and when $Q_0 = 0$ for down-counting

$$J_1 = K_1 = (C \cdot Q_0) + (\overline{C} \cdot \overline{Q_0})$$

- **MSB (Q_2):** Toggles when $Q_1Q_0 = 11$ for up-counting and when $Q_1Q_0 = 00$ for down-counting

$$J_2 = K_2 = (C \cdot Q_1 \cdot Q_0) + (\overline{C} \cdot \overline{Q_1} \cdot \overline{Q_0})$$

5.3 Operation Summary

When $C = 1$, the counter increments synchronously on each rising clock edge. When $C = 0$, the counter decrements in reverse order. Because all flip-flops share the same clock, the design avoids ripple delay and provides predictable timing behavior.

Exercise 6: Digital Clock Using Counter Components

This exercise examines the operation of the built-in counter components in Logisim by designing a digital clock that displays minutes and seconds using 7-segment LED displays.

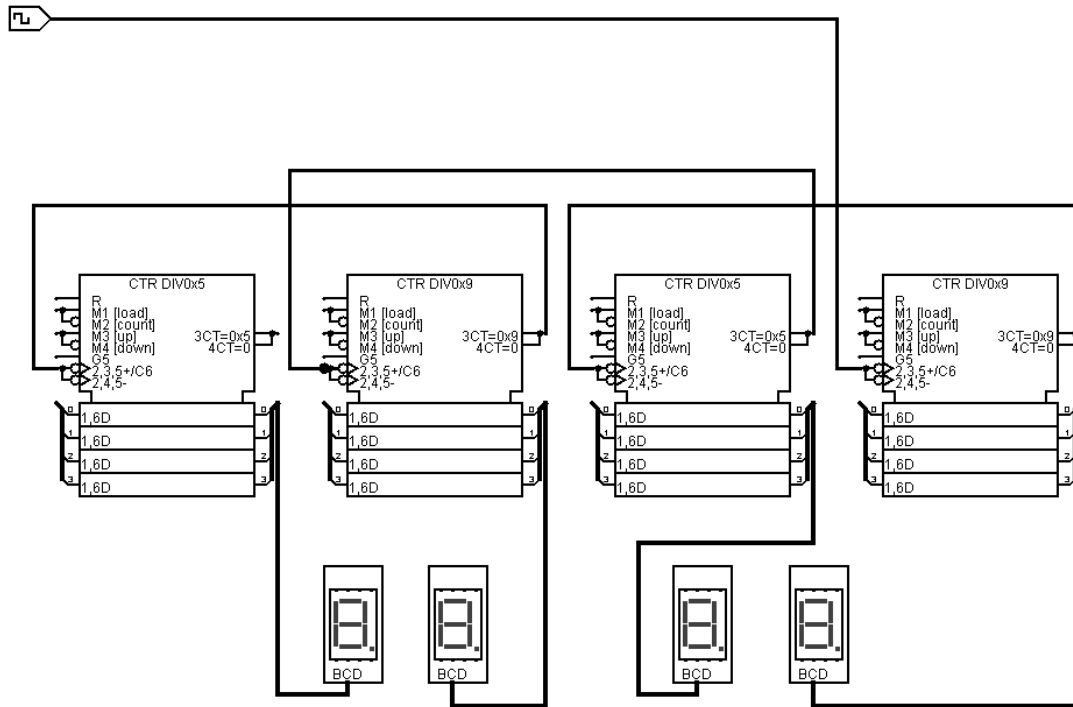


Figure 5: Digital clock implemented using Logisim counter components

6.1 Clock Structure

The digital clock is composed of four cascaded counters:

- Seconds (units): Modulo-10 counter (0–9)
- Seconds (tens): Modulo-6 counter (0–5)
- Minutes (units): Modulo-10 counter (0–9)
- Minutes (tens): Modulo-6 counter (0–5)

A 1 Hz clock signal is used to increment the seconds unit counter.

6.2 Counter Cascading

Each counter generates a carry-out signal when it reaches its maximum count. This signal is used to trigger the next higher-order counter, forming a complete time-counting system:

Seconds → Minutes

This cascading mechanism ensures accurate time progression without manual intervention.

6.3 Display Output

The output of each counter is connected to a 7-segment display configured in BCD mode. This allows the numerical values of seconds and minutes to be displayed directly without additional decoding logic.

6.4 Additional Features

Optional enhancements, such as manual time-setting using push buttons or clock multiplexing, can be added to improve usability. However, the core design successfully demonstrates the practical use of counter components in digital timekeeping applications.